

Metacognitive effort regulation in problem-solving

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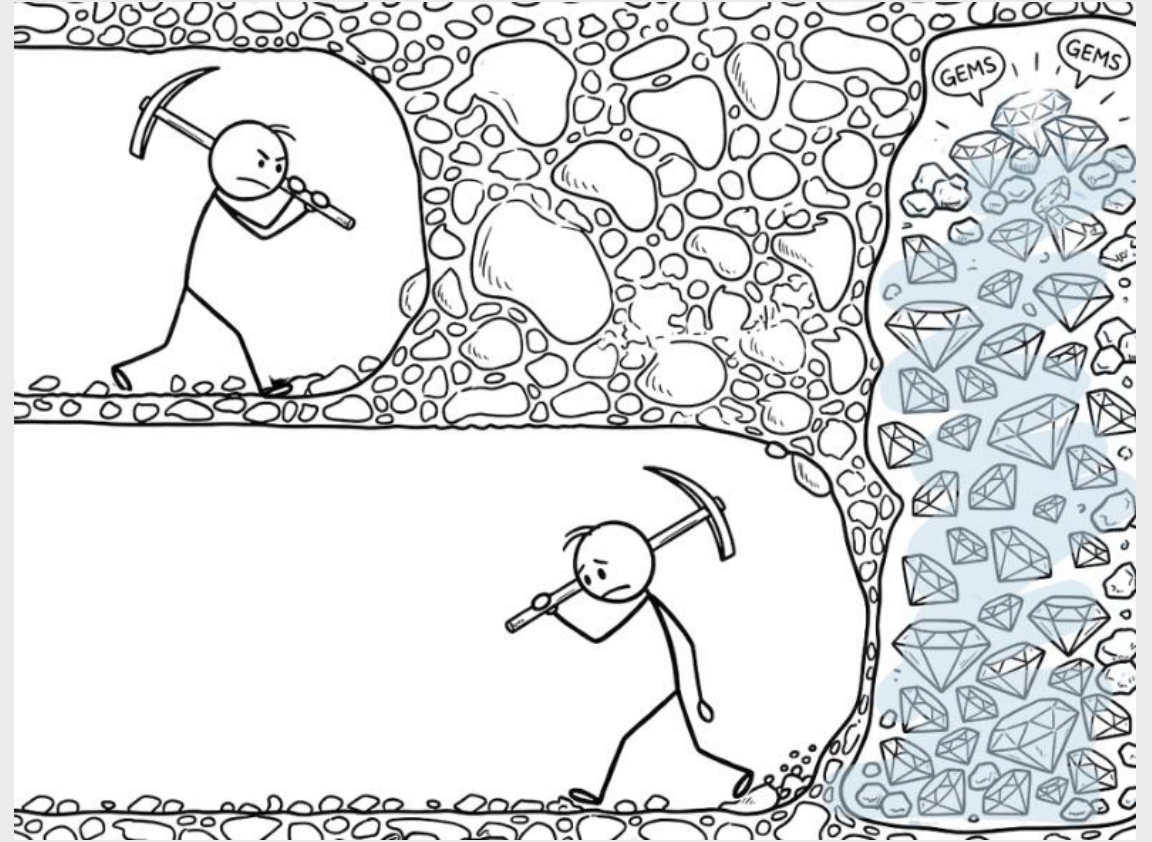
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Persistence vs. Abandonment

Persistence is often seen as key to success:

- *“Never give up! You may be closer than you think!”*

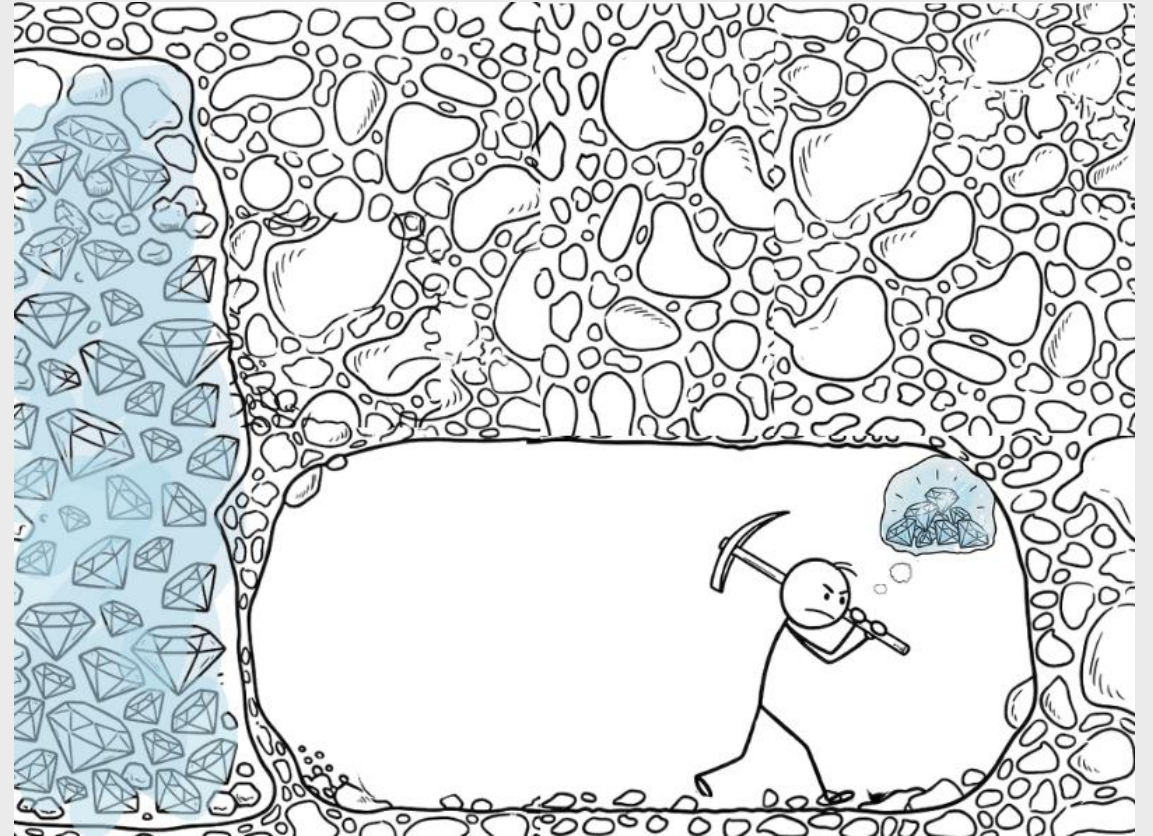


Persistence vs. Abandonment

Persistence is often seen as key to success:

- *“Never give up! You may be closer than you think!”*

But knowing when to *quit* can be just as adaptive



Persistence vs. Abandonment

Individual differences (e.g., Grit, Need for Cognition) – who persists?

Cognitive biases (e.g., sunk cost fallacy) – what distorts the decision?

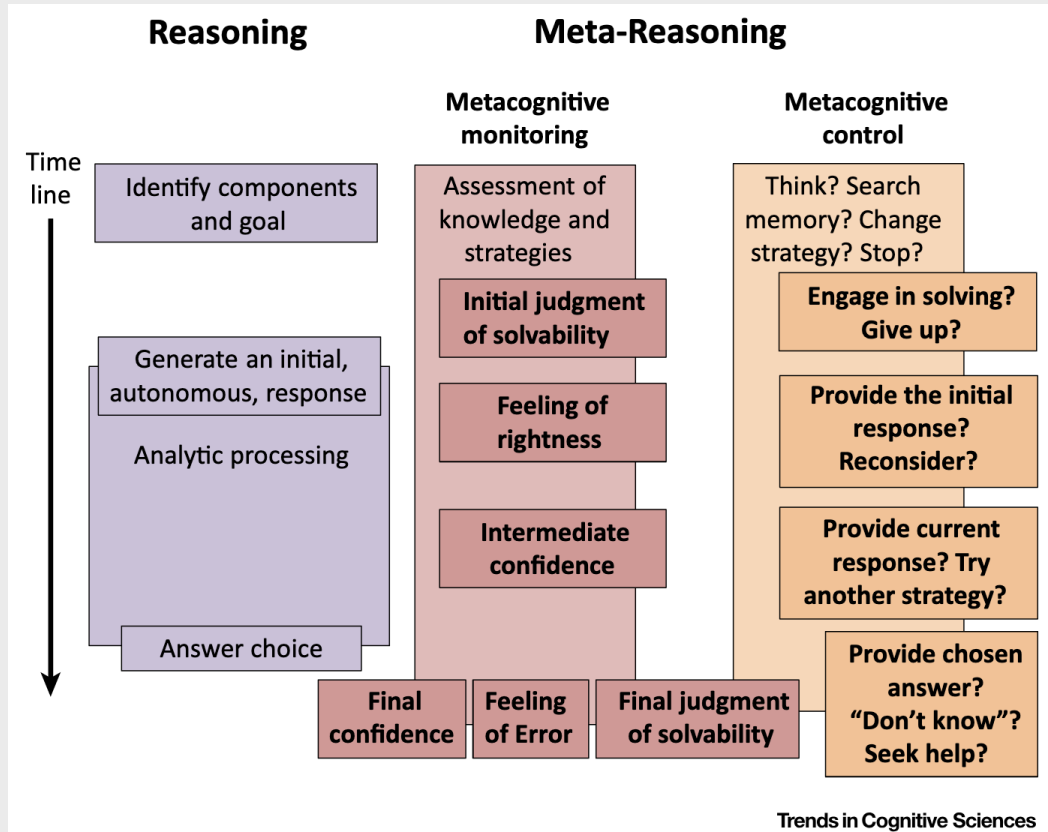
How does the decision to quit unfold during problem-solving?



**“Winners quit all the time.
They just quit the right
stuff at the right time.”**

- Seth Godin

Meta-reasoning framework



Monitoring and control of cognitive processes during problem-solving

Monitoring – assessing one's own cognitive processes (e.g., confidence)

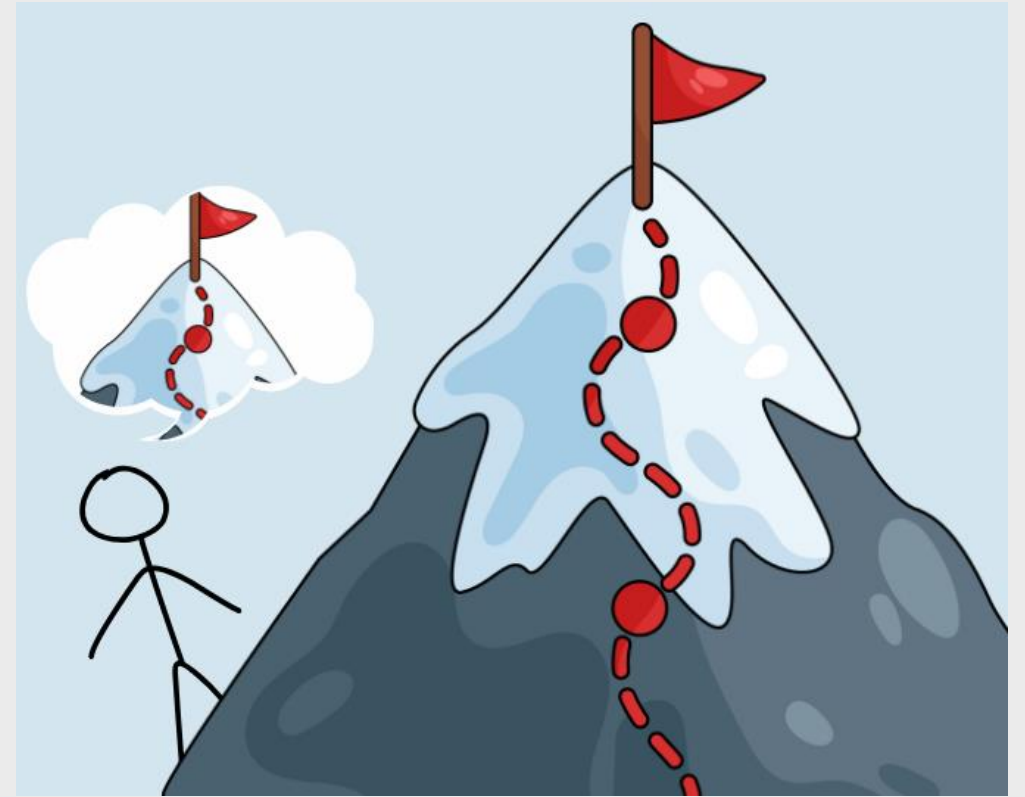
Control – effort regulation decisions in response to monitoring (e.g., rethinking, seeking help, quitting)

Disengagement as a metacognitive control decision

People abandon problems when confidence drops below a certain internal threshold

Accuracy is higher for attempted problems, suggesting strategic abandonment of low-confidence items

Response times for abandoned problems tend to be longer than for solved ones



Prospective monitoring & effort regulation



People form early impressions of problem difficulty before an attempt

Prospective monitoring judgements is associated with later solving attempts

How are these early judgements incorporated into effort regulation decisions is not yet well understood.

Anagram problem-solving

NHTGSI → INSIGHT

Variables affecting solving difficulty:

- ***Pronounceability*** – How word-like the scrambled letter string looks before solving
- ***Length***
- ***Solution SBF*** – How commonly the letter pairs in the solution word appear together in English
- ***Word frequency*** – How common the solution word is in everyday language

Stimuli selection

Anagrams generated using the **GRAMS** package in R (Newman, 2026)

Manipulations:

- Anagram pronounceability (easy vs. hard)
- Length (5 vs. 6 letter words)

PANOR RNPOA
BOLTEG TBEOGL

Controlled for # of syllables in the solution word, # of moves from a solution and # of intact bigrams in anagrams

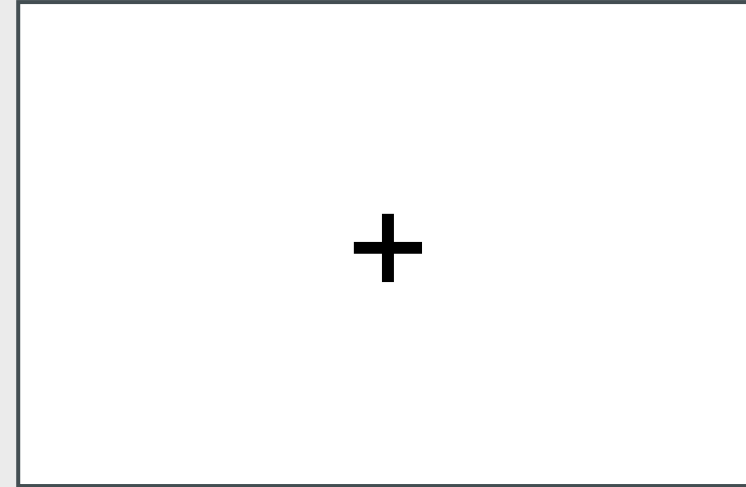
Covariates: Word frequency, solution SBF

Experiment 1

N = 100

DVs:

- Judgement of Difficulty (JOD)
- Probability of solving / abandoning
- RT before disengagement
- Fatigue (i.e. position in the task sequence)



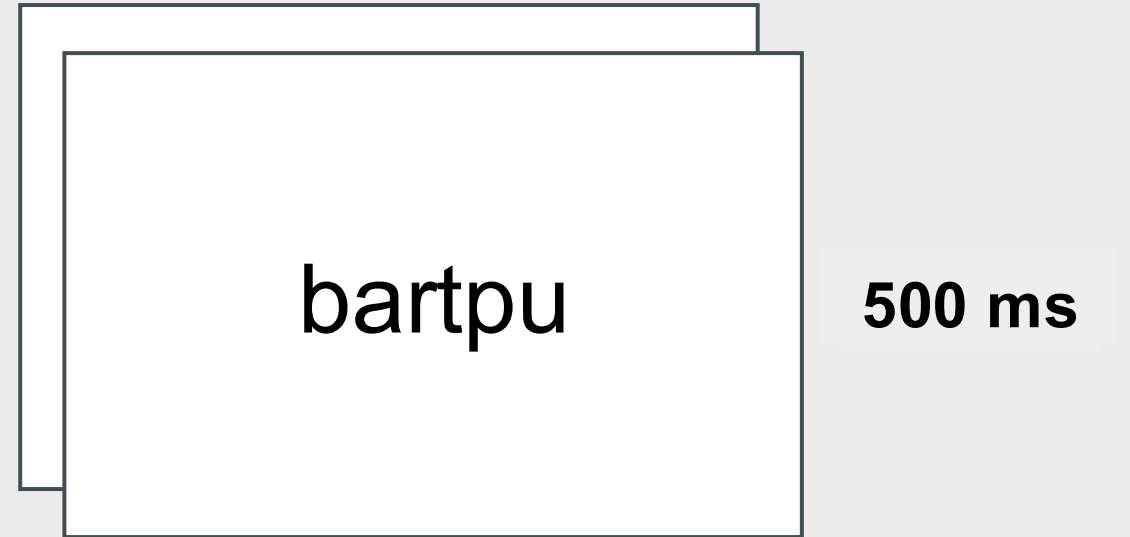
1000 ms

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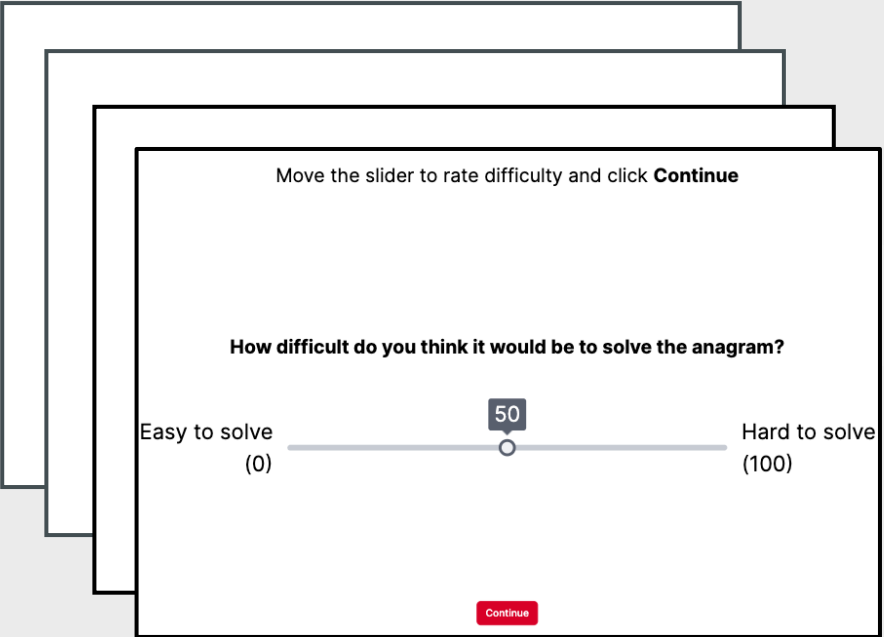


Experiment 1

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DVs:

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Move the slider to rate difficulty and click **Continue**

How difficult do you think it would be to solve the anagram?

Easy to solve (0) ————— 50 ————— Hard to solve (100)

Continue

The image shows a sequence of four overlapping rectangular boxes, representing a task sequence. The front-most box contains a slider interface for rating difficulty. At the top, it says 'Move the slider to rate difficulty and click Continue'. Below that is the question 'How difficult do you think it would be to solve the anagram?'. A horizontal slider is shown with 'Easy to solve (0)' on the left and 'Hard to solve (100)' on the right. A small box with the number '50' is positioned above the slider's handle, which is currently at the midpoint. At the bottom right of the box is a red 'Continue' button.

Experiment 1

N = 100

DVs:

- Judgement of Difficulty (JOD)
- Probability of solving / abandoning
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The diagram illustrates a sequence of overlapping rectangular screens, representing a task sequence. The topmost screen contains the following text and elements:

- Instruction: "Move the slider to rate difficulty and click **Continue**"
- Question: "Did the solution come to you during the preview?"
- Response buttons: "No" and "Yes"

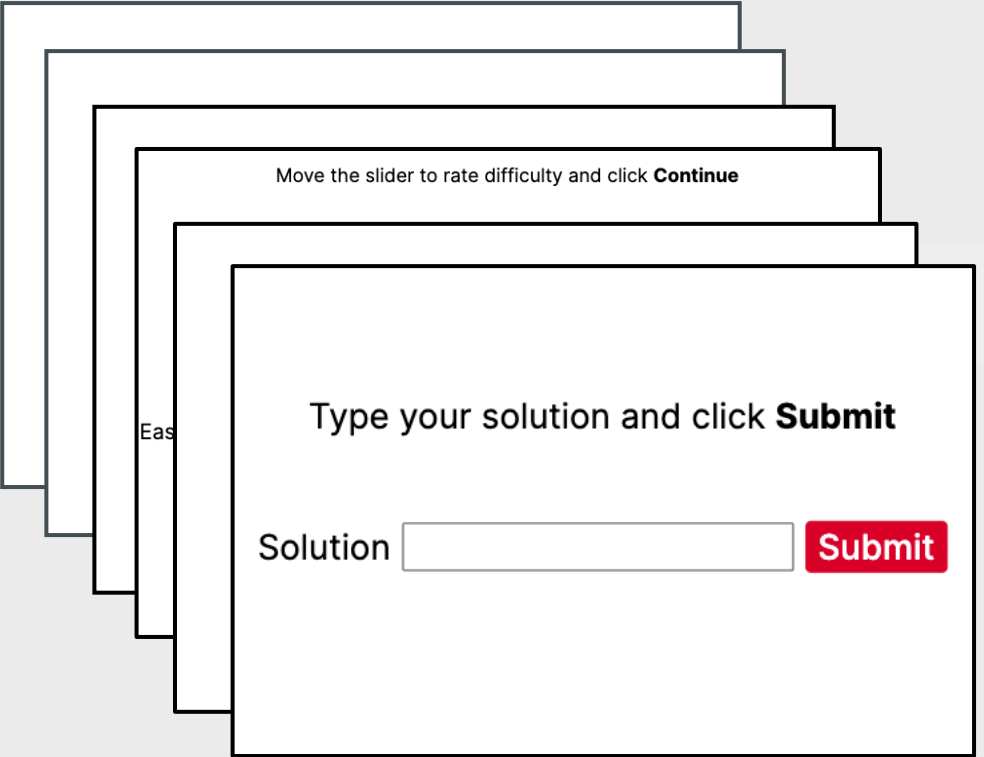
Below the question, the word "Easy" is partially visible, suggesting a difficulty level label. The overlapping nature of the screens indicates a progression through the task.

Experiment 1

N = 100

DVs:

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- Probability of solving / abandoning
- RT before disengagement
- Fatigue (i.e. position in the task sequence)



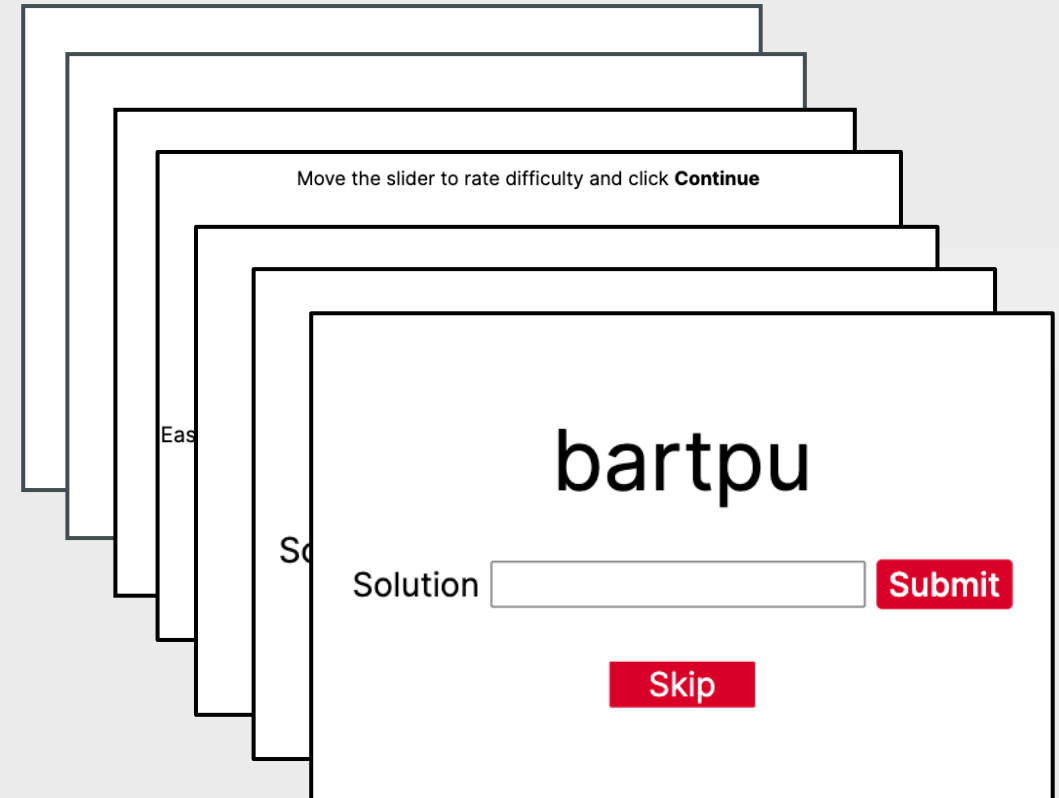
The diagram illustrates a sequence of task screens. The top-most screen displays the instruction: "Move the slider to rate difficulty and click **Continue**". Below it, a second screen is partially visible, showing the word "Easy" on the left. The bottom-most screen is the most prominent and contains the instruction: "Type your solution and click **Submit**". This screen features a text input field labeled "Solution" and a red button labeled "Submit".

Experiment 1

N = 100

DVs:

- Judgement of Difficulty (JOD)
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Analysis strategy

LMMs & GLMM using lme4 package in R

$$(1) \text{ JOD} \sim \text{Pronounceability} \times \text{Length} + \text{WF} + \text{SBF}$$

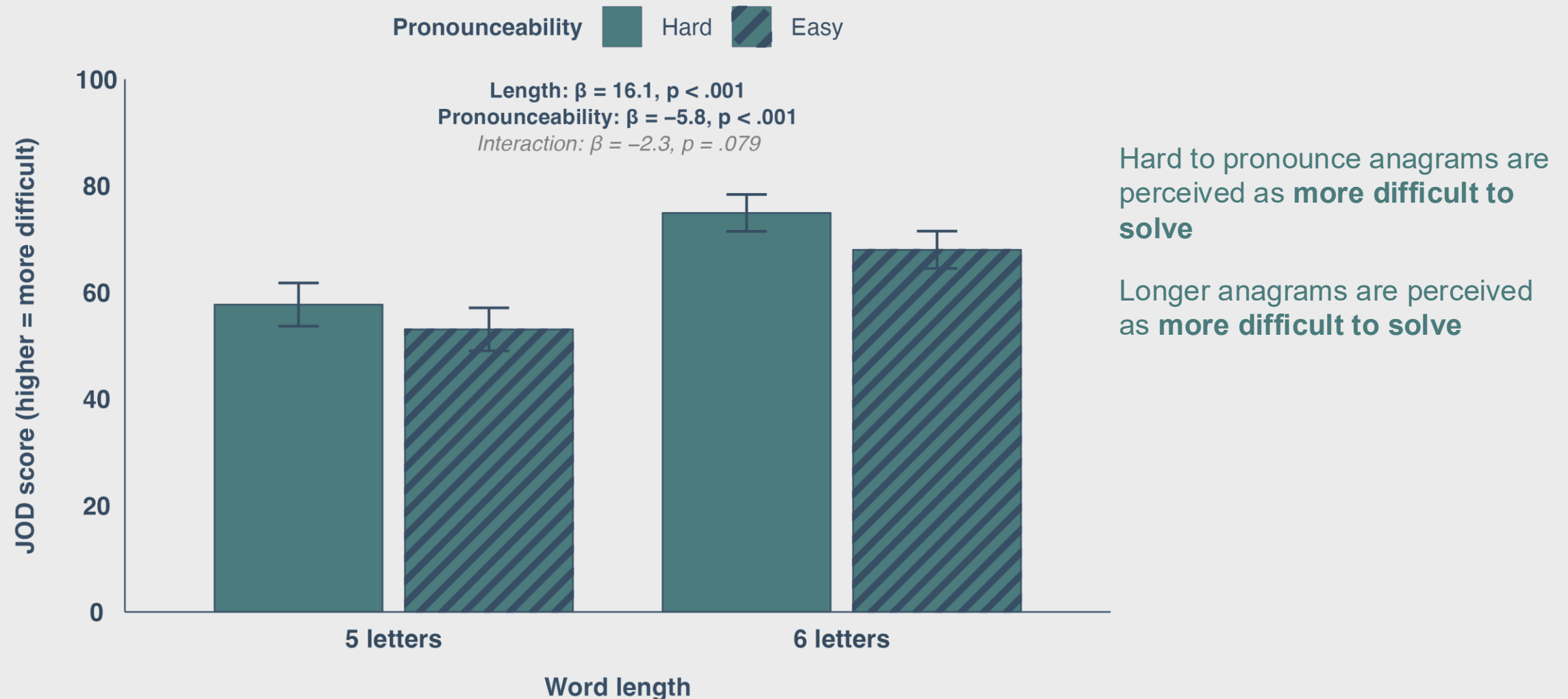
$$(2) \text{ P(Solved)} \sim \text{Pronounceability} \times \text{Length} + \text{JOD} + \text{Fatigue} + \text{WF} + \text{SBF}$$

$$(3) \text{ log(RT)} \sim \text{Pronounceability} \times \text{Length} + \text{JOD} + \text{Fatigue} + \text{WF} + \text{SBF}$$

Maximal approach justified by the design:

- Random intercepts for participants and items
- Random slopes for participants where estimable

How do people form an initial assessment of difficulty?

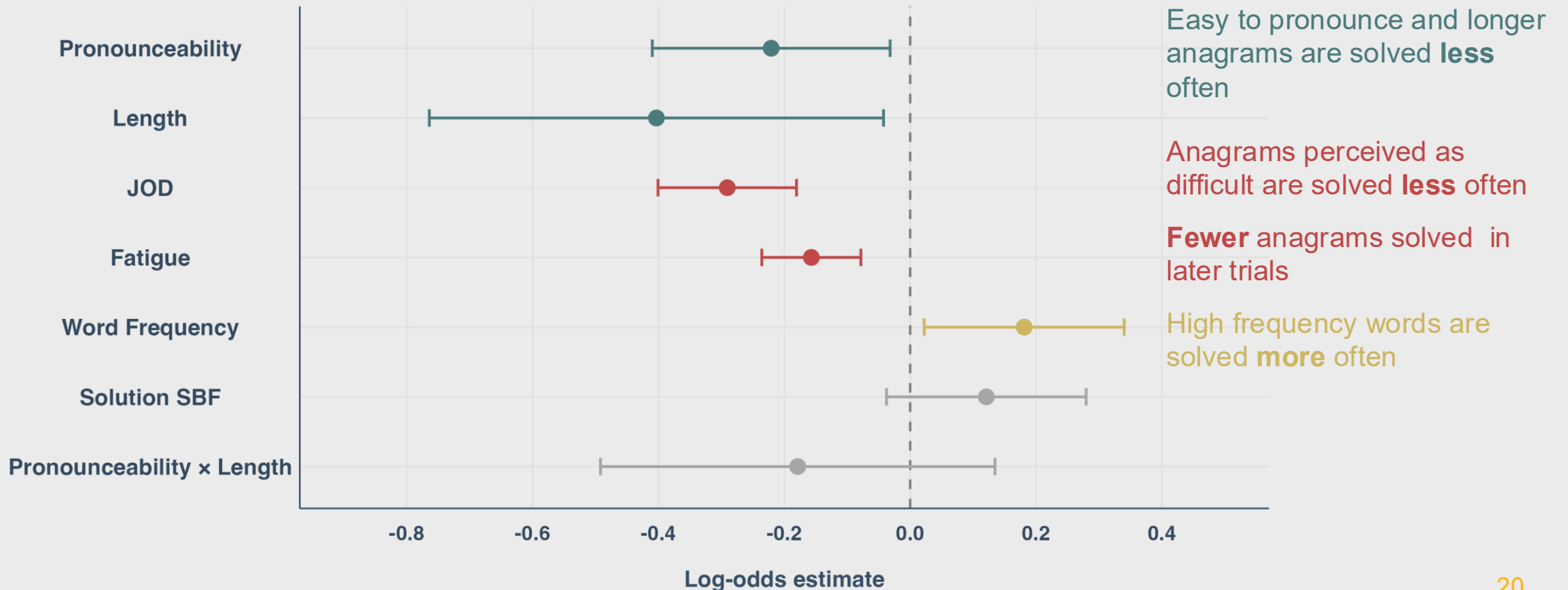


Note. Error bars represent 95% confidence intervals around model-estimated means.

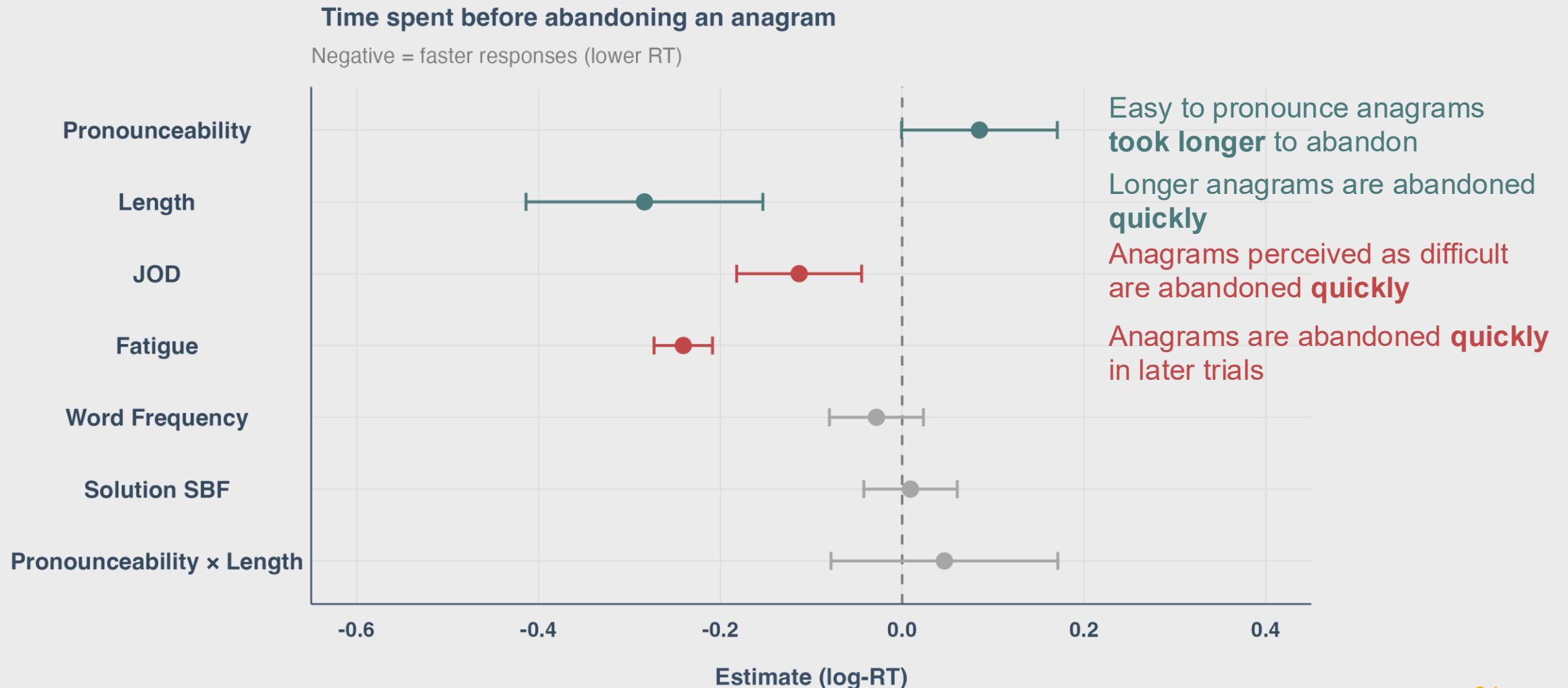
What predicts whether people solve or abandon an anagram?

Probability of solving or abandoning an anagram

Negative = reduces solving probability

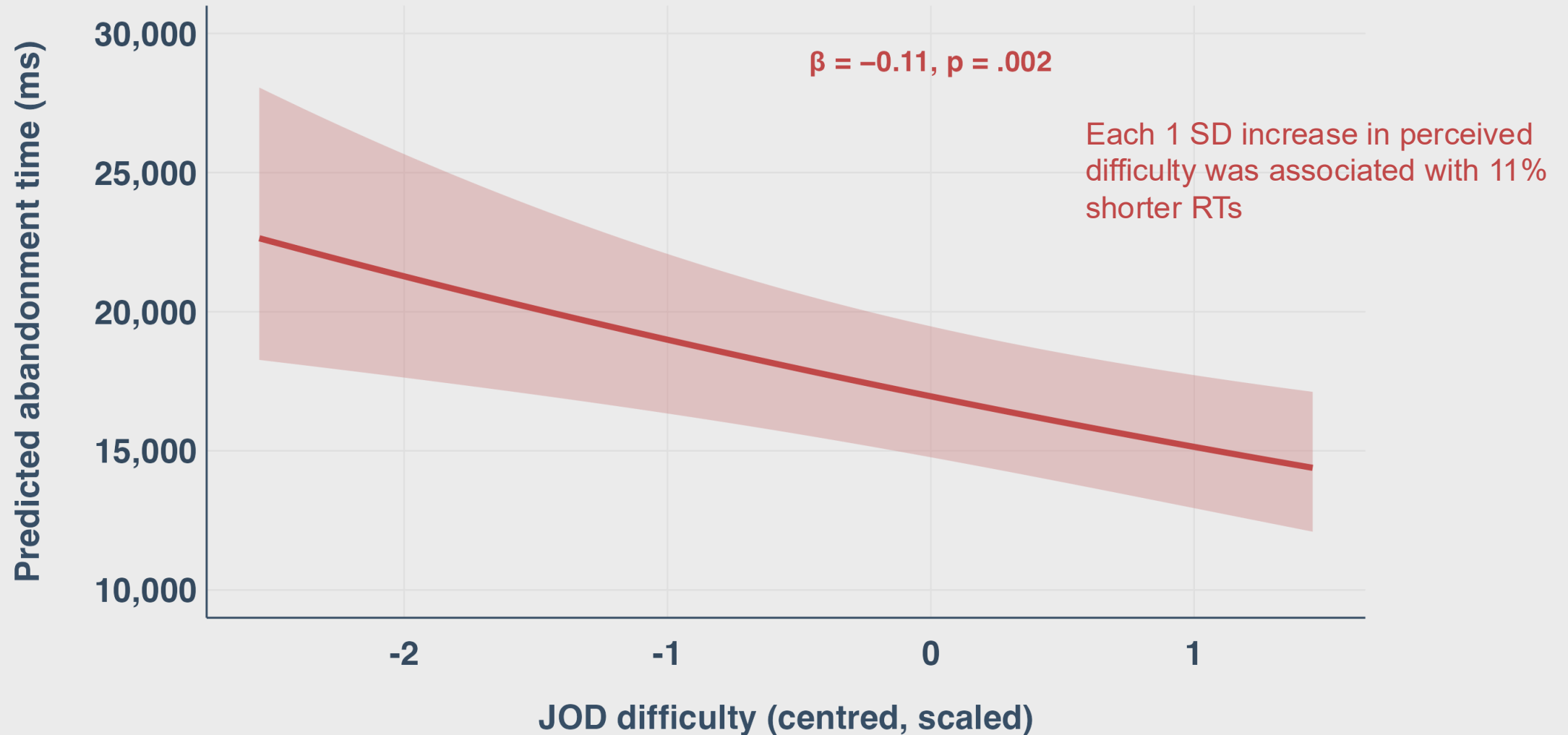


What predicts how long people take before abandoning an anagram?



Note. Model is based on trials where participants chose to abandon the anagram

Initial assessment of difficulty & effort regulation



Note. β coefficients are on the log scale

Conclusions

People use salient problem features to form prospective difficulty judgements

This initial perception of difficulty predicts how long people persist before abandoning problems

Other variables may play a role:

Self-efficacy signals – monitoring incremental progress

Monitoring feedback – performance on previous trials

Switch costs – cognitive cost of reorienting to a new problem

GRAMS package in R for computing **psycholinguistic metrics** and generating **anagram stimuli** by Ian Newman:

